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An explainable AI framework for multimodal student performance prediction and institutional decision support at PSAU

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A B S T R A C T

Improving student success and supporting evidence-based decision-making in higher education depend on early detection of academically at-risk students. An Explainable Artificial Intelligence (XAI) framework is proposed in this paper for multimodal student performance prediction to support institutional decision-making at Prince Sattam bin Abdulaziz University (PSAU). The framework was developed and evaluated using a publicly available educational dataset collected from a higher education institution, integrating data from the Student Information System (SIS), Moodle Learning Management System (LMS), and the eDify mobile learning application. The proposed framework combines comprehensive data preprocessing, Mutual Information-based feature selection, Optuna-based hyperparameter optimization, and an optimized CatBoost classifier. Using repeated stratified 5-fold cross-validation, the model's performance was assessed and compared with statistical learning methods, conventional machine learning algorithms, homogeneous ensemble methods, and heterogeneous ensemble approaches. The proposed model demonstrated superior predictive efficacy, attaining an Accuracy of 92.0%, Precision of 92.7%, Recall of 97.9%, F1-score of 95.2%, ROC-AUC of 0.913, PR-AUC of 0.974, Balanced Accuracy of 0.824, and Matthews Correlation Coefficient of 0.726, significantly exceeding all rival models according to the Wilcoxon signed-rank test ($p < 0.001$). Mutual Information feature selection further improved predictive performance while reducing model complexity. Explainability was achieved through CatBoost feature importance and SHAP analyses, which consistently identified assessment outcomes and online learning engagement variables as the most influential predictors. The proposed framework provides a transferable blueprint for implementing explainable early-warning systems at Prince Sattam bin Abdulaziz University and other higher education institutions to support timely interventions, personalized learning support, and evidence-based institutional decision-making.

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1. Introduction

The forecasting of student academic achievement has emerged as a significant area of inquiry in Learning Analytics (LA) and Educational Data Mining (EDM), as higher education institutions increasingly seek data-driven approaches to pinpoint at-risk students and deliver prompt academic support. Accurate prediction models allow instructors to track student advancement, allocate academic support more effectively, and improve student retention and success rates. With the widespread adoption of Student Information Systems (SISs) and Learning Management Systems (LMSs), and digital learning platforms, educational institutions now generate large volumes of heterogeneous data describing students' academic records, learning behaviors, and online engagement, creating new opportunities for intelligent predictive analytics (Hasan et al., 2021; Nawang et al., 2021). Traditional statistical approaches have been widely applied to student performance prediction; however, they often assume linear relationships and predefined functional forms that are incapable of adequately encapsulating the intricate dynamics among academic, behavioral, and online learning variables. Consequently, machine learning methodologies have gained significant traction due to their ability to model nonlinear relationships and automatically learn complex decision

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boundaries from educational data (Hasan et al., 2020; Jokhan et al., 2022). Recent research has shown that ensemble learning techniques, such as Gradient Boosting, Extreme Gradient Boosting (XGBoost), and Random Forest, generally outperform conventional statistical models by effectively exploiting nonlinear relationships and feature interactions (Cosar, 2024; Guevara-Reyes et al., 2025).

Among modern ensemble learning algorithms, Categorical Boosting (CatBoost) has attracted increasing attention because it is specifically designed to process categorical variables efficiently while reducing prediction bias through ordered boosting and robust encoding techniques (Prokhorenkova et al., 2018). These characteristics make CatBoost particularly suitable for educational datasets, which typically contain numerous categorical academic indicators alongside behavioral variables. Nevertheless, many previous studies employ default hyperparameter settings or conventional tuning strategies, potentially limiting predictive performance and model generalization. Concurrently, Explainable Artificial Intelligence (XAI) has emerged as an essential component of educational prediction systems. Although highly accurate machine learning models can effectively identify at-risk students, their practical adoption is often constrained by their limited interpretability. Educational decision-makers require transparent explanations that justify model predictions and identify the academic and behavioral factors influencing student outcomes. Consequently, SHapley Additive exPlanations (SHAP) have become increasingly popular for improving transparency and supporting evidence-based educational interventions (Gori et al., 2025; Hidayatulloh et al., 2026). Notwithstanding the significant advancements made in forecasting student performance, some critical research voids persist. First, relatively few studies have systematically optimized CatBoost using modern hyperparameter optimization frameworks such as Optuna, despite their demonstrated ability to improve predictive performance and generalization (Akiba et al., 2019). Second, prior studies have primarily focused on either academic records or LMS data individually, with comparatively limited attention devoted to multimodal educational datasets that integrate information from student information systems, online learning platforms, and video learning analytics. Third, although SHAP has become increasingly popular, many studies use explainability only as a supplementary visualization technique instead of being integrated into a comprehensive explainable prediction framework. Finally, previous research frequently relies on a limited number of evaluation metrics, with relatively few studies combining multiple complementary classification measures with statistical significance testing to rigorously compare competing prediction models.

To address these gaps, this study proposes an Optuna-optimized CatBoost classifier for predicting student academic performance using a multimodal educational dataset that integrates multimodal data from SIS, Moodle Learning Management System, and eDify mobile learning application (Hasan et al., 2021). The proposed framework incorporates comprehensive data preprocessing, Mutual Information-based feature selection, systematic hyperparameter optimization, and Explainable Artificial Intelligence through SHAP analysis. The optimized model is evaluated against statistical learning models, traditional machine learning models, as well as homogeneous and heterogeneous ensemble-based approaches using repeated stratified cross-validation, multiple complementary classification metrics, and the Wilcoxon signed-rank test. Beyond its methodological contributions, the proposed framework has practical relevance for institutional decision support in higher education. Although the model is developed and evaluated using a publicly available multimodal educational dataset collected from a higher education institution in the Sultanate of Oman, the framework is designed to be institution-independent and can be adapted to other universities with comparable data infrastructures. In particular, the proposed Explainable AI framework aligns with the ongoing digital transformation initiatives at Prince Sattam bin Abdulaziz University (PSAU), where LMSs, SISs, and LA are increasingly utilized to support evidence-based academic decision-making. By combining accurate prediction with transparent SHAP-based explanations, the framework provides a foundation for developing intelligent early-warning systems that assist academic advisors, instructors, and university administrators in identifying students requiring timely academic support.

This research advances the educational data mining literature through four major contributions. First, it demonstrates that systematic Optuna-based hyperparameter optimization substantially improves the predictive capability of CatBoost compared with a broad range of competing statistical and machine learning approaches. Second, it shows that Mutual Information feature selection effectively reduces model complexity while improving predictive performance. Third, it provides a comprehensive evaluation framework by combining multiple classification performance metrics with statistical significance testing. Finally, it enhances model transparency through SHAP-based explainability, providing interpretable insights into how academic assessments and online learning behaviors jointly influence student performance predictions. Collectively, these contributions present a comprehensive, accurate, and interpretable framework that supports the timely identification of academically at-risk students and promotes data-driven decision-making in higher education.

2. Literature Review

2.1. Student Performance Prediction and Machine Learning Approaches

The prediction of student academic performance has become a central research theme within Learning Analytics (LA) and Educational Data Mining (EDM), particularly as higher education institutions increasingly rely on data-driven decision-making to enhance student outcomes and institutional effectiveness. Early work in this domain emphasized the importance of identifying key predictors of student success and failure using structured academic and demographic data. For instance, Daud

et al. (2017) aimed to improve prediction accuracy by incorporating students' personal and financial attributes into classification models. Their methodology relied on combining learning analytics techniques with discriminative and generative models, and their findings demonstrated that enriched feature sets significantly outperform traditional models that rely solely on academic indicators. Similarly, Pardo et al. (2017) adopted a mixed-method approach integrating self-regulated learning indicators with behavioral data from online learning environments. Their results showed that combining psychological and behavioral factors provides a more comprehensive explanation of academic performance, highlighting the multidimensional nature of learning processes.

Systematic evidence further supports the importance of feature diversity in predictive modeling. Nawang et al. (2021), through a systematic literature review, identified six major feature categories commonly used in student performance prediction: demographic, academic, e-learning, social interaction, institutional, and prior education variables. Their findings revealed that academic performance indicators remain the most frequently used predictors, but combining multiple feature categories significantly enhances model robustness and predictive accuracy. This insight underscores the importance of integrating heterogeneous data sources, particularly in modern digital learning environments where students interact with multiple platforms simultaneously. Advances in machine learning have enabled researchers to investigate a broad range of predictive algorithms for improving the accuracy of student academic performance prediction. Conventional models, including Decision Trees (DT) and Support Vector Machines (SVM), have been extensively adopted because of their computational efficiency, simplicity, and interpretability; however, more sophisticated models such as Gradient Boosting and Random Forest methods have exhibited superior predictive capabilities. Hasan et al. (2020) conducted an empirical study employing data mining technique in conjunction with video learning analytics on data collected from 772 students. Their methodology involved applying multiple classification algorithms alongside methods, including principal component analysis and information gain. Among the evaluated models, Random Forest achieved the highest accuracy (88.3%), indicating its superior ability to capture the complex and nonlinear characteristics of educational data. Similarly, Jokhan et al. (2022) introduced an AI-based early warning system that employed a Random Forest algorithm trained on data from an institutional Early Warning System (EWS). trained on data from an institutional Early Warning System (EWS). Their model achieved an impressive accuracy of 97.03%, underscoring the strong predictive capability of ensemble-based approaches in identifying at-risk students and supporting timely interventions.

Gradient boosting techniques, particularly XGBoost, have also gained prominence attributed to their effectiveness in modeling complex feature interactions and handle missing data efficiently. Cosar (2024) developed a predictive model using XGBoost, achieving an R^2 score of 0.819, with key predictors including exam scores, participation rates, and demographic variables. Building on this, Guevara-Reyes et al. (2025) employed both XGBoost and Random Forest models on a large dataset of 50,000 student records. Their study utilized advanced preprocessing techniques, hyperparameter tuning, and cross-validation, resulting in an R^2 of 0.91 and a significant reduction in mean squared error. Importantly, their findings revealed that a small subset of variables, such as socioeconomic status, access to technology, and prior academic performance, explained a considerable proportion of the variability in student outcomes. These results underscore the potential of ML models not only to predict performance but also to uncover critical factors influencing student success.

Despite substantial progress in predictive modeling, model interpretability remains a persistent challenge. High-performing algorithms, including Random Forest and XGBoost, are frequently characterized as black-box models, limiting stakeholders' ability to interpret the decision-making process underlying their predictions. This limitation poses significant challenges for institutional decision-making, where transparency and accountability are essential. As a result, there has been growing interest in integrating XAI techniques into student performance prediction models.

2.2. Explainable AI, Ensemble Learning, and Research Gaps

To overcome the interpretability limitations of advanced machine learning models, Explainable Artificial Intelligence (XAI) has emerged as a powerful paradigm. By providing interpretable insights into the prediction process, XAI enables educators and administrators to better understand, trust, and effectively utilize model outputs in educational decision-making. A notable contribution in this area is the study by Gori et al. (2025), who developed an XAI-based framework for student performance prediction that achieved 93% accuracy. The study evaluated XGBoost, Random Forest, and neural network models, and integrated SHAP and LIME to interpret model predictions. Their findings identified attendance, assignment performance, and quiz scores as key determinants of academic performance, demonstrating the value of explainability in identifying actionable factors. Similarly, Hidayatulloh et al. (2026) adopted a structured experimental approach to develop an explainable prediction model using Random Forest and XGBoost algorithms. Their methodology included data preprocessing, model training, evaluation, and the application of XAI techniques for interpretability. The results showed that both models achieved high accuracy levels (above 89%), while SHAP and LIME analyses identified key predictors such as previous academic performance, failures, and absenteeism. These findings emphasize the importance of combining predictive accuracy with interpretability to support effective interventions. George et al. (2025) further reinforced this perspective by demonstrating that midterm exams and assignment scores are critical determinants of student performance, with XAI techniques providing clear insights into feature importance and relationships. Beyond individual prediction models, recent research has explored the integration of XAI into broader learning analytics systems and decision-support frameworks. Afzaal et al. (2021) developed a data-driven feedback and intelligent action recommendation system that combines explainable machine learning

with a user-friendly dashboard. Their study demonstrated that providing transparent insights into student performance can significantly enhance self-regulated learning and improve academic outcomes. Similarly, Alonso & Casalino (2019) introduced an explainable system capable of generating graphical and textual explanations, enabling educators to better understand and communicate predictive results. These studies highlight the potential of XAI to transform predictive analytics into actionable insights that support informed educational decision-making.

Ensemble learning has emerged as a robust strategy for improving prediction accuracy by leveraging the complementary strengths of multiple machine learning models. Ahmed et al. (2025) developed a voting-based ensemble regression framework by integrating several ML algorithms and incorporates SHAP and LIME for interpretability. The findings revealed that the ensemble model consistently outperformed the standalone models, yielding an R^2 of 0.989 and providing valuable insights into key predictors such as extracurricular activities, sleep patterns, and preparation levels. Similarly, Hoq et al. (2023) introduced a stacked ensemble learning framework to predict student performance in programming courses, demonstrating superior predictive power relative to individual models. Their approach also utilized SHAP to explain model predictions, highlighting the importance of integrating explainability into ensemble frameworks. More advanced frameworks have further extended the capabilities of ensemble learning. Kirar et al. (2026) proposed a cognitive-generative decision intelligence framework that combines ensemble techniques with natural language generation to produce human-readable explanations. Their model achieved exceptionally high performance metrics, including 98.9% accuracy, while maintaining interpretability. This approach represents a significant step toward fully automated, explainable decision-support systems in education. However, despite these advancements, most ensemble models focus primarily on combining similar types of algorithms, such as tree-based models, with limited integration of fundamentally different learning paradigms.

Systematic reviews provide additional insights into the current state of research and highlight several persistent limitations. Choi et al. (2025) found that while SHAP is widely used for global feature importance analysis, there is limited emphasis on personalized explanations and actionable insights. Alamri & Alharbi (2021) identified a strong reliance on traditional models such as Decision Trees, beside insufficient adoption of advanced XAI techniques and limited integration of e-learning behavioral data. Furthermore, Chitti et al. (2020) emphasized the need for interpretable models, noting that many high-performing approaches lack transparency and are therefore difficult to implement in real-world educational settings. Another important limitation is the underutilization of multimodal datasets. Many studies rely on a single data source, such as academic records or LMS activity, without fully leveraging the rich information available across multiple platforms. The dataset proposed integrates SIS data, Moodle interactions, and mobile learning data from the eDify application, represents a valuable yet underexplored resource. This dataset captures a comprehensive view of student behavior, including academic performance, online engagement, and video interaction patterns, offering significant potential for improving predictive models. Despite the substantial progress in machine learning for student performance prediction, several important research gaps remain. First, although numerous studies have compared a variety of machine learning algorithms, relatively few have investigated whether systematic hyperparameter optimization can substantially improve the predictive capability of advanced boosting algorithms such as CatBoost. Many existing studies employ default parameter settings or conventional tuning strategies, which may limit model performance and generalizability. Second, although CatBoost has demonstrated strong predictive capability in educational data as it can effectively learn complex nonlinear relationships and interdependencies among predictor variables, its application to multimodal educational datasets remains relatively limited. Most previous studies have focused on academic records or LMS data individually, rather than integrating multiple sources of educational information that jointly capture students' academic background, learning behavior, and online engagement.

Third, although XAI has attracted increasing attention in educational data mining, many studies rely primarily on global feature importance measures or employ SHAP solely as a post hoc visualization technique. Consequently, limited research has combined optimized gradient boosting models with comprehensive explainability analyses that provide both global and local interpretations of prediction outcomes. Finally, previous studies have predominantly evaluated prediction models using a limited number of performance measures or simple accuracy comparisons. More rigorous evaluations employing multiple complementary classification metrics together with statistical significance testing remain comparatively scarce, reducing confidence in the reported superiority of proposed models.

To address these research gaps, this study proposes an Optuna-optimized CatBoost classifier for predicting student academic performance using a multimodal educational dataset that integrates information from SIS, Moodle learning management system, and the eDify mobile learning application. The proposed framework combines comprehensive data preprocessing with feature selection using Mutual Information (MI) to retain the most informative predictors while reducing model complexity. Unlike previous studies that frequently employ default hyperparameter settings, the proposed CatBoost model is systematically optimized using the Optuna framework with the TPE sampler, allowing efficient exploration of the hyperparameter space to maximize predictive performance. The optimized model is evaluated against a comprehensive set of statistical learning methods, conventional machine learning algorithms, homogeneous ensemble models, and heterogeneous ensemble approaches using repeated stratified cross-validation, multiple classification performance metrics, and the Wilcoxon signed-rank test. To enhance transparency and support educational decision-making, the study incorporates XAI through SHAP analysis. Both global feature importance and SHAP summary plots are employed to identify the most influential predictors and explain how individual feature values contribute to the prediction of student success or failure. The proposed

framework seamlessly integrates accurate predictive modeling with XAI, offering educators with actionable insights into the academic and behavioral factors influencing student performance.

3. Proposed Explainable Prediction Framework

3.1. General Framework

This research proposes an explainable machine learning framework for predicting student academic performance using an Optuna-optimized CatBoost classifier. As shown in Fig. 1, the framework begins with a comprehensive data preprocessing phase, including missing value treatment, numerical feature preparation, and categorical variable encoding to ensure data quality and consistency. Mutual Information (MI) is then employed to select the most informative predictors. The selected features are used to train a CatBoost classifier, whose hyperparameters are optimized using Optuna with the TPE sampler. The suggested model is subsequently contrasted with statistical models, individual machine learning algorithms, and homogeneous and heterogeneous ensemble methods. Model performance is evaluated using repeated stratified cross-validation and multiple classification metrics, while the Wilcoxon signed-rank test is applied to assess the statistical significance of performance differences. Finally, Feature importance analysis, SHAP summary and SHAP bar plots are used to interpret the model by identifying the most influential predictors of student academic performance.

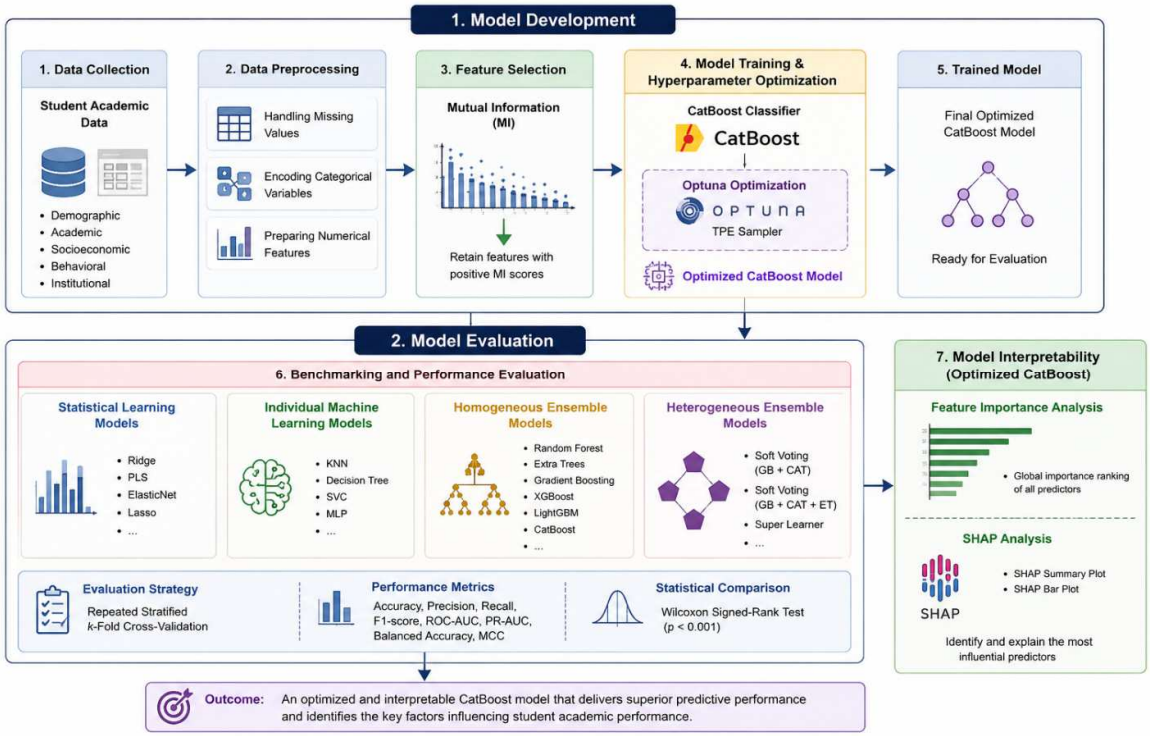


Fig. 1. General framework of the proposed study.

3.2. Phase 1: Data Preparation

Step 1: Dataset Description

The current research employs a publicly available educational dataset developed by Hasan et al. (2021). The dataset was collected from a higher education institution in the Sultanate of Oman and integrates data from three institutional platforms: the Student Information System (SIS), the Moodle Learning Management System (LMS), and the eDify mobile learning application. It covers five academic modules delivered between Spring 2017 and Spring 2021 and comprises 326 student records with 40 explanatory variables. These variables capture students' academic characteristics, learning behaviors, assessment performance, and online engagement.

After data inspection, one incomplete record was excluded, resulting in a final dataset of 325 student observations. Following feature engineering and feature selection, 24 informative predictors were retained for model development. The target variable is the students' final academic performance, formulated as a binary classification problem (Pass/Fail). The resulting dataset provides comprehensive information on students' academic records, assessment outcomes, and online learning activities, making it suitable for educational data mining and machine learning-based prediction of student performance.

Step 2: Data Preprocessing

Prior to model development, the dataset was examined for duplicate records and missing values to ensure data quality. Duplicate observations were removed, and no missing values were detected; consequently, no imputation procedures were required. The target variable was encoded into a binary format, where Fail = 0 and Pass = 1, to facilitate binary classification. The predictor variables were categorized into 15 categorical variables and 4 numerical variables according to their measurement scales (Table 1). To ensure appropriate preprocessing, numerical variables (*Played, Paused, Likes,* and *Segment*) were standardized using StandardScaler, while categorical variables were transformed using One-Hot Encoding.

Table 1
Variables Measurement

Variable	Description
CGPA	Student's cumulative grade point average represented by academic performance categories (e.g., *Fair, Adequate, Good, Very Good, Excellent*).
AttemptCount	Categorized number of attempts made by the student to complete the course (e.g., *Low, Medium, High*).
RemoteStudent	Indicates whether the student is enrolled in remote learning mode (*Yes* or *No*).
Probation	Indicates whether the student is under academic probation (*Yes* or *No*).
HighRisk	Indicates whether the registered course has historically exhibited a high failure rate (*Yes* or *No*).
TermExceeded	Indicates whether the student has exceeded the expected duration of the academic program (*Yes* or *No*).
AtRisk	Indicates whether the student has previously failed two or more courses (*Yes* or *No*).
AtRiskSSC	Indicates whether the Student Success Center has identified the student as academically at risk (*Yes* or *No*).
OtherModules	Level of the student's enrollment in other courses during the current semester (e.g., *Low, Medium, High*).
PlagiarismHistory	Indicates whether the student has a previous history of academic integrity violations (*Yes* or *No*).
CW1	Student's achievement level in the first coursework assessment.
CW2	Student's achievement level in the second coursework assessment.
ESE	Student's achievement level in the end-semester examination.
Online C	Category representing the student's level of interaction with online course content.
Online O	Category representing the student's participation in online learning activities.
Played	Number of learning resources or videos accessed by the student through the learning management system.
Paused	Number of pauses recorded while viewing online learning materials.
Likes	Number of positive interactions (likes) recorded during online learning activities.
Segment	Number of learning content segments viewed or completed by the student.
Result	Final student academic outcome used as the target variable, classified as *Pass* or *Fail*.

Step 3: Feature Selection

Feature selection was performed after data preprocessing using Mutual Information (MI), a filter-based feature selection method that evaluates the statistical dependence between individual predictors and the target variable, effectively detecting both linear and nonlinear relationships. Because it operates independently of linear assumptions, unlike traditional correlation metrics, and functions prior to model training, MI offers a computationally efficient approach well-suited for high-dimensional data (Liu & Motani, 2025). Following one-hot encoding, the feature space expanded to 45 encoded features. To reduce dimensionality while preserving the most informative predictors, only features with positive MI scores (MI > 0) were retained, reducing the feature set to 24 features. This process eliminated non-informative variables, improved computational efficiency, and enhanced the quality of the input features for subsequent machine learning model development.

3.3. Phase 2: Model Development

Step 1: Development of the Suggested Prediction Model

In this phase, a supervised ensemble framework is developed for predicting students' academic performance using an Optuna-optimized CatBoost classifier. The objective is to accurately distinguish between students who pass and those who fail by capturing the complex nonlinear relationships and interactions among academic characteristics, assessment outcomes, and online learning behaviors. Unlike conventional statistical methods that rely on predefined functional forms, ensemble learning combines the outputs of multiple models to generate a more accurate and reliable prediction than that achieved by an individual model. through the aggregation of complementary learning patterns (Kunapuli, 2023; Mohammed & Kora, 2023). Numerous studies have demonstrated that ensemble methods consistently outperform individual learning algorithms across a wide range of prediction tasks (Ali, 2026e, 2026d, 2026c). Ensemble learning algorithms are broadly categorized into homogeneous and heterogeneous methods. Homogeneous ensemble methods employ a single learning algorithm to construct multiple base learners, including bagging methods as Extra Trees (ET) and Random Forests (RF), and boosting methods such as Gradient Boosting (GB), Extreme Gradient Boosting (XGBoost), Light Gradient Boosting Machine (LightGBM), and Categorical Boosting (CatBoost). In contrast, heterogeneous ensembles combine predictions from different learning algorithms using techniques such as Soft Voting and Super Learner (Ali, 2026a, 2026b; Mienye et al., 2022). The present study adopts CatBoost as the proposed model because of its ability to effectively model complex nonlinear relationships while efficiently handling categorical variables through ordered boosting and advanced encoding strategies. To comprehensively evaluate its effectiveness, the proposed Optuna-optimized CatBoost model is benchmarked against four groups of competing methods: (i) statistical learning models (Ridge, PLS, ElasticNet, and Lasso), (ii) conventional machine learning algorithms (KNN,

Decision Tree, SVC, and MLP), (iii) homogeneous ensemble learning methods (Gradient Boosting, LightGBM, XGBoost, Extra Trees, Random Forest, and the baseline CatBoost), and (iv) heterogeneous ensemble learning methods (Soft Voting and Super Learner). This comprehensive benchmarking framework enables a rigorous assessment of whether the proposed model delivers superior predictive performance across different categories of predictive algorithms. This section describes the implementation of the CatBoost algorithm in conjunction with the Optuna-based hyperparameter tuning strategy used in this research.

(a) Categorical Boosting (CatBoost) Classifier

Introduced by Dorogush et al. (2018), Categorical Boosting (CatBoost) is a state-of-the-art gradient boosting algorithm that incrementally builds an ensemble of decision trees, with every tree trained to reduce the prediction errors of preceding trees. In contrast to conventional gradient boosting methods, CatBoost incorporates ordered boosting and efficient encoding strategies for categorical variables, thereby reducing prediction bias, mitigating overfitting, and improving model generalization (Prokhorenkova et al., 2018). Furthermore, CatBoost employs a balanced level-wise tree-growing strategy and random permutations during training, together with an improved leaf-value estimation procedure, which enhances training efficiency and addresses the gradient bias issue commonly associated with traditional boosting methods (Lee et al., 2023).
CatBoost is particularly well suited for educational data because student performance is determined by complex nonlinear interactions among academic characteristics, assessment outcomes, and online learning behaviors. Its capability to automatically model nonlinear associations and higher-order feature interactions with minimal feature engineering makes it an effective approach for predicting students' academic performance. Consequently, CatBoost was adopted as the core predictive model in the present study and further enhanced through Optuna-based hyperparameter optimization to maximize predictive performance and generalization ability.

(b) Optuna-Based Hyperparameter Optimization

The predictive performance of CatBoost depends considerably on its hyperparameter configuration. Instead of using manually determined hyperparameter values or exhaustive grid search, the current study employs the Optuna optimization framework to automatically identify an optimal combination of hyperparameters. Optuna provides an efficient framework for hyperparameter optimization by applying adaptive sampling strategies to identify promising regions within the hyperparameter search space while minimizing unnecessary model evaluations (Akiba et al., 2019; Ali, 2026a; Ali & Yahia, 2025). Hyperparameter optimization was performed using the TPE sampler implemented in Optuna. The optimization process aimed to maximize the mean ROC–AUC score estimated through stratified five-fold cross-validation. To ensure reproducibility, 50 optimization trials were performed using a fixed random seed of 42. During each trial, CatBoost hyperparameter such as learning rate, tree depth, L2 leaf regularization, random strength, bagging temperature, border count, the number of boosting iterations, and tree growth policy, were sampled from predefined search spaces (Table 2) and evaluated using cross-validation. For the *Lossguide* growth policy, the maximum number of leaves was optimized as an additional hyperparameter. The hyperparameter combination that maximized the cross-validated ROC–AUC score was adopted to construct the final proposed model.

Using Optuna provides several advantages over conventional tuning approaches. By adaptively concentrating the search on promising regions of the hyperparameter space, it improves predictive performance while reducing computational cost compared with exhaustive search methods (Akiba et al., 2019; Ali & Alaskar, 2026). The optimized hyperparameter search space and the final selected configuration are presented in Table 2.

Table 2
Hyperparameter search space and optimal configuration of the proposed CatBoost model.

Hyperparameter	Search space	Optimal value
Iterations	200–600	410
Tree depth	4–8	8
Learning rate	0.01–0.30 (log scale)	0.230
L2 leaf regularization	1–20	7.246
Random strength	0–10	1.049
Bagging temperature	0–10	6.028
Border count	32–255	89
Grow policy	{SymmetricTree, Depthwise, Lossguide}	Depthwise
Maximum leaves*	31–128	N/A

*Optimized only when the Lossguide growth policy was selected.

Fig. 2 illustrates the conceptual workflow of the Optuna optimization process. The workflow begins with defining the CatBoost hyperparameter search space, followed by iterative hyperparameter sampling using the TPE algorithm. Each sampled configuration is evaluated through stratified cross-validation, after which the optimization study records the resulting performance and guides subsequent trials toward more promising regions of the search space. After the optimization converges, the best-performing hyperparameter configuration is used to construct the final optimized CatBoost classifier,

which is subsequently evaluated using multiple classification metrics and interpreted through feature importance analysis and SHAP-based explainable artificial intelligence techniques.

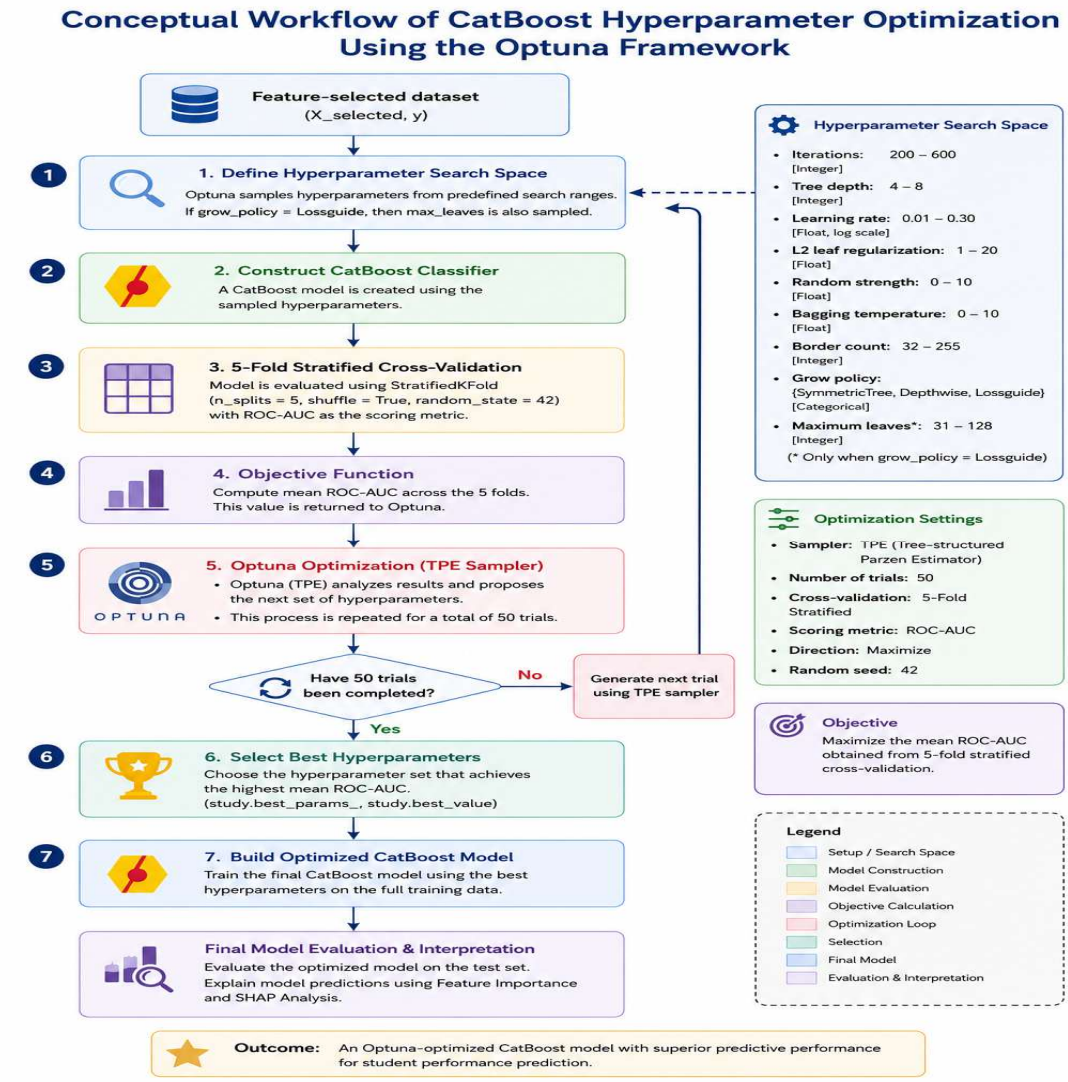


Fig. 2. Conceptual workflow of the Optuna optimization process.

Step 2: Comparative Performance Evaluation

The proposed Optuna-optimized CatBoost classifier was subjected to a comparative performance assessment against a diverse set of benchmark models using stratified 5-fold cross-validation. The comparison included four categories of predictive methods: (i) statistical learning models (Ridge, Partial Least Squares (PLS), Elastic Net, and Lasso); (ii) individual machine learning models (Multi-Layer Perceptron (MLP), Decision Tree, Support Vector Classifier (SVC), and K-Nearest Neighbors (KNN)); (iii) homogeneous ensemble models (Gradient Boosting, LightGBM, XGBoost, Extra Trees, Random Forest, and the baseline CatBoost classifier); and (iv) heterogeneous ensemble models, including Soft Voting (GB + CAT), Soft Voting (GB + CAT + ET), and the Super Learner ensemble (GB + CAT + ET with logistic regression meta learner).

To provide a comprehensive assessment of predictive performance, eight complementary classification metrics were employed: Accuracy, Precision, Recall, F1-score, ROC–AUC, Precision–Recall Area Under the Curve (PR-AUC), Balanced Accuracy, and Matthews Correlation Coefficient (MCC). Accuracy represents the proportion of correctly classified samples, whereas Precision measures the proportion of predicted positive instances that are correctly identified, and Recall measures the proportion of actual positive instances that are successfully detected. The F1-score provides a balanced measure of Precision and Recall. ROC-AUC evaluates the model's ability to discriminate between classes across different decision thresholds, while PR-AUC is particularly informative for datasets with class imbalance. Balanced Accuracy accounts for unequal class distributions by averaging class-specific recall, and MCC provides a robust overall measure of classification quality by considering all elements of the confusion matrix. Collectively, these metrics offer a holistic evaluation of predictive

performance and facilitate robust comparisons among competing classification models (Foody, 2023; Chicco & Jurman, 2023).

Step 3: Statistical Significance Test

The Wilcoxon signed-rank test was conducted to determine the statistical significance of the observed performance differences between the proposed model and the competing models. The test compared the fold-wise ROC-AUC values obtained from the proposed Optuna-optimized CatBoost classifier with those of each benchmark model across the stratified 5-fold cross-validation procedure. The Wilcoxon signed-rank test is a non-parametric paired statistical method that does not rely on the assumption of a normal data distribution and is therefore well suited for comparing machine learning models evaluated on the same data partitions (Demšar, 2006; Sziklai et al., 2021). The statistical comparisons were conducted separately for statistical learning models, individual machine learning models, homogeneous ensemble methods, and heterogeneous ensemble methods. For each comparison, the Wilcoxon test statistic and the corresponding significance level were reported to determine whether differences in predictive performance were statistically significant.

Step 4: Model Interpretation

To enhance the interpretability and transparency of the suggested model, XAI techniques was incorporated through SHapley Additive exPlanations (SHAP). The TreeExplainer algorithm, specifically developed for tree-based ensemble models, was employed to compute SHAP values (Lundberg & Lee, 2017). The optimized CatBoost classifier was trained using the optimal hyperparameter configuration obtained through Optuna, and SHAP analysis was subsequently applied to quantify the contribution of each feature to the model's predictions.

Global feature importance was assessed by ranking predictors according to their mean absolute SHAP values, thereby identifying the variables with the greatest influence on student performance prediction. In addition, SHAP summary plots were used to illustrate both the magnitude and direction of each feature's contribution across all observations, providing insights into how academic performance indicators and online learning behaviors influenced the model's predictions. These explainability techniques improve the interpretability of the proposed classifier while supporting the practical understanding of the factors associated with student academic performance.

4. Experimental Results

The empirical results of the proposed framework are presented in this section. The analysis begins with a descriptive summary of the study variables and subsequently examines the predictive performance of the proposed model against benchmark statistical and machine learning approaches. Next, feature importance analysis identifies the most influential predictors of student performance. Finally, SHAP-based explainability analysis is performed to provide transparent and interpretable explanations of the model's predictions, thereby supporting trustworthy educational decision-making.

4.1. Descriptive Analysis

Table 3 and Table 4 exhibit the descriptive statistics of the numerical variables and the distribution of the categorical variables used to predict students' academic performance. The numerical engagement variables (Tables 3) indicate moderate levels of interaction with the online learning environment. On average, students played learning resources 2.13 times and paused videos 2.19 times, while the mean values of Likes (0.997) and Segment (1.48) suggest relatively limited interaction with online learning materials. Among these variables, Paused exhibited the largest variability (SD = 2.605), indicating substantial differences in students' viewing behaviors. All numerical variables displayed positive skewness, implying that most students demonstrated relatively low levels of online engagement, whereas a smaller proportion exhibited considerably higher activity. In particular, Paused showed the highest skewness (1.638) and kurtosis (3.576), indicating a distribution characterized by several extreme observations.

Table 4 summarizes the distributions of the categorical variables. The majority of students had a low attempt count (92.6%), were not remote learners (99.7%), were not on academic probation (96.6%), and were not classified as high risk (93.2%), suggesting that most students were enrolled under normal academic conditions. Nevertheless, approximately 23.7% of the students had previously been identified as academically at risk, highlighting the presence of a meaningful subgroup with prior academic difficulties. Academic integrity issues were rare, with only 0.3% of students having a recorded plagiarism history. Assessment outcomes revealed substantial variation in student performance. Nearly half of the students failed the first coursework assessment (49.8%) and more than half failed the second coursework assessment (53.2%), whereas the end-semester examination exhibited a more balanced distribution, with 38.8% of students achieving an Adequate grade and 31.7% receiving a Fail grade. These findings indicate that continuous assessment components were generally more challenging than the final examination. Regarding online learning behaviors, the majority of students were classified as having Good, Very Good, or Excellent levels of participation in both online content access and online learning activities, suggesting relatively active engagement with the learning management system.

Overall, the descriptive statistics indicate considerable heterogeneity in students' academic performance and learning behaviors. The positively skewed distributions of the numerical engagement variables, together with the substantial variability observed across several academic indicators, suggest the presence of complex and potentially nonlinear relationships between student characteristics and academic outcomes. These distributional characteristics support the use of an optimized CatBoost classifier, which is capable of modeling nonlinear interactions among heterogeneous predictor variables without requiring restrictive distributional assumptions.

Table 3
Descriptive statistics of numerical variables.

Variable	Mean	Median	Std. Dev.	Min	Max	Skewness	Kurtosis
Played	2.126	1.000	1.760	0	8	1.236	0.872
Paused	2.191	1.000	2.605	0	13	1.638	3.576
Likes	0.997	1.000	0.967	0	3	0.724	-0.431
Segment	1.480	0.000	1.877	0	7	1.011	-0.105

Table 4
Distribution of categorical variables.

Variable	Categories (%)
CGPA	Adequate (36.9), Fair (28.3), Good (17.5), Very Good (10.8), Excellent (5.5), Poor (0.9)
Attempt Count	Low (92.6), Medium (6.2), High (1.2)
Remote Student	No (99.7), Yes (0.3)
Probation	No (96.6), Yes (3.4)
High Risk	No (93.2), Yes (6.8)
Term Exceeded	No (98.2), Yes (1.8)
At Risk	No (76.3), Yes (23.7)
At Risk SSC	No (95.4), Yes (4.6)
Other Modules	High (52.0), Medium (30.8), Low (17.2)
Plagiarism History	Low (99.7), Medium (0.3)
CW1	Fail (49.8), Adequate (16.9), Fair (11.7), Good (10.8), Very Good (7.7), Excellent (3.1)
CW2	Fail (53.2), Adequate (14.2), Very Good (12.0), Fair (10.2), Good (5.5), Excellent (4.9)
Final Exam (ESE)	Adequate (38.8), Fail (31.7), Fair (12.3), Good (7.7), Very Good (6.5), Excellent (3.1)
Online Content	Good (34.2), Very Good (23.4), Excellent (22.8), Fair (9.8), Poor (7.4), Adequate (2.5)
Online Activities	Good (25.5), Very Good (24.0), Excellent (21.2), Fair (12.6), Poor (11.1), Adequate (5.5)

4.2. Feature Selection Results

To examine the effectiveness of the suggested feature selection method, the performance of the optimized CatBoost classifier was compared using the complete encoded feature set (45 features) and the subset of informative features selected through Mutual Information (24 features). Table 5 summarizes the comparative results obtained from repeated stratified five-fold cross-validation repeated 10 times, while statistical significance was assessed using the Wilcoxon signed-rank test. The results demonstrate that feature selection consistently improved the prediction performance of the proposed model. Using the selected feature subset increased Accuracy from 91.42% to 92.15% ($p = 0.0238$), ROC-AUC from 0.8615 to 0.8848 ($p < 0.001$), Balanced Accuracy from 0.8102 to 0.8309 ($p < 0.001$), and Matthews Correlation Coefficient (MCC) from 0.7023 to 0.7321 ($p < 0.001$). Although the F1-score also increased from 0.9487 to 0.9528, the difference was not statistically significant ($p = 0.0913$). Overall, the findings indicate that eliminating non-informative features enhances the discriminative ability of the CatBoost classifier while simultaneously reducing the dimensionality of the feature space from 45 to 24 variables. The statistically significant improvements observed for ROC-AUC, Balanced Accuracy, and MCC further suggest that the reduced feature set enables the model to generalize more effectively by retaining the most informative predictors and mitigating the influence of irrelevant variables. Consequently, the Mutual Information-selected feature subset was adopted for all subsequent model development and comparative analyses.

Table 5
Effect of Mutual Information feature selection on CatBoost performance.

Metric	All Features (45)	Selected Features (24)	Gain	Wilcoxon p-value
Accuracy	0.9142	0.9215	+0.0073	0.0238*
F1-score	0.9487	0.9528	+0.0041	0.0913
ROC-AUC	0.8615	0.8848	+0.0233	<0.001***
Balanced Accuracy	0.8102	0.8309	+0.0207	<0.001***
MCC	0.7023	0.7321	+0.0298	<0.001***

Note: *Bold values indicate the superior performance. Statistical significance was evaluated by the Wilcoxon signed-rank test on the fold-wise scores achieved from repeated stratified five-fold cross-validation repeated 10 times (50 evaluation folds).* $p < 0.05$ (*); $p < 0.001$ (***).

4.3. Comparison with Baseline Models

Tables 4–7 and Fig. 3 compare the predictive performance of the proposed Optuna-optimized CatBoost classifier with four categories of benchmark methods, including statistical learning models, conventional machine learning algorithms, homogeneous ensemble methods, and heterogeneous ensemble learning approaches. Model performance was evaluated using multiple complementary classification metrics, while statistical significance was assessed using the Wilcoxon signed-rank

test based on the fold-wise ROC-AUC values obtained from stratified 5-fold cross-validation. Overall, the proposed model demonstrated the strongest discriminative performance among all competing approaches, achieving a ROC-AUC of 0.913, PR-AUC of 0.974, F1-score of 0.952, Balanced Accuracy of 0.824, and MCC of 0.726. Although several benchmark models produced comparable values for Accuracy and F1-score, none matched the proposed model in ROC-AUC, indicating its superior ability to distinguish between students who pass and those who fail across different classification thresholds.

As shown in Table 6, the proposed model substantially outperformed all statistical learning models. Ridge and PLS achieved moderate discriminative performance (ROC-AUC = 0.790 and 0.789, respectively), whereas ElasticNet and Lasso yielded ROC-AUC values of 0.500, indicating performance equivalent to random classification. Relative to these statistical approaches, the proposed model improved ROC-AUC by approximately 12–41 percentage points. Statistical significance testing using the Wilcoxon signed-rank test verified that the observed performance improvements were significant in every comparison ($p < 0.001$). Table 7 presents the comparison with conventional machine learning algorithms. Among these methods, Decision Tree achieved the highest ROC-AUC (0.829), followed closely by MLP (0.828), SVC (0.812), and KNN (0.792). While SVC produced competitive Accuracy (0.917) and F1-score (0.950), its ROC-AUC remained more than 10 percentage points below that of the proposed model. The statistically significant Wilcoxon test results ($p < 0.001$) indicate that the observed improvements in discriminative performance are significant.

The comparison with homogeneous ensemble methods is reported in Table 8. Among the benchmark ensemble models, Gradient Boosting achieved the highest ROC-AUC (0.872), followed by XGBoost (0.869), Extra Trees (0.866), LightGBM (0.855), Baseline CatBoost (0.853), and Random Forest (0.852). Notably, although the baseline CatBoost attained the highest Accuracy (0.924), F1-score (0.955), and MCC (0.739), its ROC-AUC remained substantially lower than that of the optimized CatBoost classifier. This finding highlights that hyperparameter optimization primarily enhanced the model's discriminative capability rather than merely improving threshold-dependent performance measures. The Wilcoxon signed-rank test further demonstrated that the observed improvements were statistically significant over all homogeneous ensemble methods ($p < 0.001$). Table 9 compares the proposed classifier with heterogeneous ensemble learning approaches. Soft Voting (GB + CAT + ET), Super Learner, and Soft Voting (GB + CAT) achieved ROC-AUC values ranging from 0.865 to 0.869, together with competitive Accuracy and F1-score values. Nevertheless, none of these ensemble combinations exceeded the ROC-AUC of the optimized CatBoost model. The Wilcoxon signed-rank test consistently demonstrated statistically significant differences favoring the proposed model across all heterogeneous ensemble methods ($p < 0.001$), suggesting that systematic hyperparameter optimization of a single CatBoost classifier was more effective than combining multiple learners for this prediction task.

Table 6
Comparison with statistical learning models

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Balanced Accuracy	MCC	Wilcoxon W
Optimized CatBoost	0.920	0.927	0.979	0.952	**0.913**	**0.974**	0.824	0.726	,
Ridge	0.678	0.916	0.664	0.768	0.790	0.912	0.701	0.324	0.0***
PLS	0.878	0.879	0.985	0.929	0.789	0.909	0.705	0.554	0.0***
ElasticNet	0.809	0.809	1.000	0.895	0.500	0.809	0.500	0.000	0.0***
Lasso	0.809	0.809	1.000	0.895	0.500	0.809	0.500	0.000	0.0***

*** indicates statistical significance at $p < 0.001$ based on the Wilcoxon signed-rank test comparing the fold-wise ROC-AUC values of each baseline model with the proposed optimized CatBoost model.

Table 7
Comparison of the proposed optimized CatBoost with individual machine learning models.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Balanced Accuracy	MCC	Wilcoxon W
Optimized CatBoost	**0.920**	**0.927**	**0.979**	**0.952**	**0.913**	**0.974**	**0.824**	**0.726**	,
SVC	0.917	0.920	0.984	0.950	0.812	0.925	0.808	0.711	1.0***
Decision Tree	0.889	0.932	0.931	0.931	0.829	0.927	0.821	0.647	3.0***
MLP	0.897	0.918	0.960	0.938	0.828	0.923	0.796	0.647	0.0***
KNN	0.869	0.878	0.973	0.923	0.792	0.914	0.699	0.513	0.0***

*** indicates statistical significance at $p < 0.001$ based on the Wilcoxon signed-rank test comparing the fold-wise ROC-AUC values of each baseline model with the proposed optimized CatBoost model.

Table 8
Comparison of the proposed optimized CatBoost with homogeneous ensemble models

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC	Balanced Accuracy	MCC	Wilcoxon W
Optimized CatBoost	**0.920**	**0.927**	**0.979**	**0.952**	**0.913**	**0.974**	**0.824**	**0.726**	,
Gradient Boosting	0.910	0.927	0.966	0.946	0.872	0.951	0.821	0.696	115.5***
XGBoost	0.915	0.925	0.974	0.949	0.869	0.953	0.819	0.708	34.0***
Extra Trees	0.914	0.920	0.979	0.949	0.866	0.945	0.808	0.703	10.0***
LightGBM	0.918	0.925	0.979	0.951	0.855	0.945	0.820	0.719	41.5***
Baseline CatBoost	0.924	0.926	0.986	0.955	0.853	0.942	0.824	0.739	0.0***
Random Forest	0.914	0.913	0.988	0.949	0.852	0.939	0.793	0.699	2.0***

*** indicates statistical significance at $p < 0.001$ based on the Wilcoxon signed-rank test comparing the fold-wise ROC-AUC values of each baseline model with the proposed optimized CatBoost model.

Table 9
Comparison of the proposed optimized CatBoost with heterogeneous ensemble learning methods.

Model	Accuracy	Precision	Recall	F1	ROC-AUC	Balanced Accuracy	MCC	Wilcoxon W
Optimized CatBoost	**0.9200**	**0.9267**	**0.9790**	**0.9520**	**0.9128**	**0.8241**	**0.7259**	,
Soft Voting (GB + CAT + ET)	0.9228	0.9272	0.9821	0.9537	0.8689	0.8265	0.7348	20.5***
Super Learner	0.9197	0.9211	0.9855	0.9521	0.8664	0.8128	0.7219	15.0***
Soft Voting (GB + CAT)	0.9209	0.9275	0.9794	0.9525	0.8649	0.8261	0.7290	17.0***

*** indicates statistical significance at $p < 0.001$ based on the Wilcoxon signed-rank test comparing the fold-wise ROC-AUC values of each baseline model with the proposed optimized CatBoost model.

Fig. 3 visually summarizes the ROC-AUC performance across the four categories of benchmark models. The proposed optimized CatBoost consistently achieved the highest ROC-AUC in every comparison group, demonstrating its robustness across diverse classes of predictive algorithms. Collectively, these findings indicate that the proposed Optuna-optimized CatBoost framework provides superior discriminative performance for student academic performance prediction, outperforming statistical learning models, conventional machine learning algorithms, homogeneous ensemble methods, and heterogeneous ensemble learning approaches.

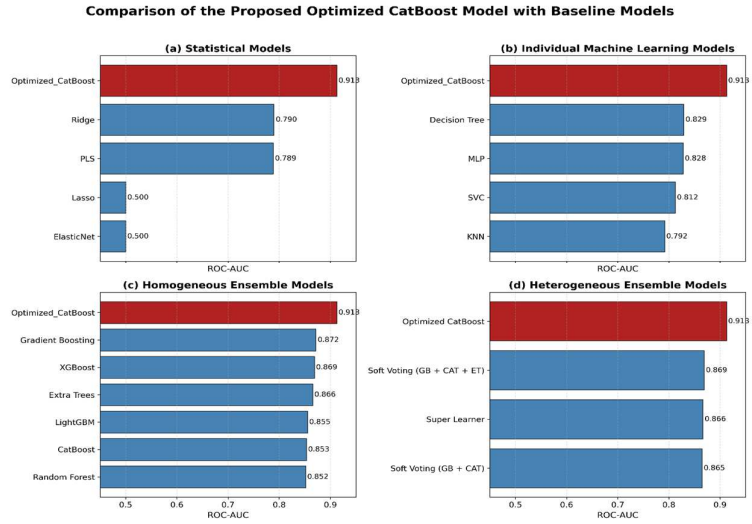


Fig. 3. Comparative performance of the Optimized CatBoost model and (a) statistical models, (b) individual machine learning models, (c) homogeneous ensemble models, and (d) heterogeneous ensemble learning methods based on ROC-AUC.

4.4. Feature Importance Analysis

Fig. 4 presents the relative importance of the 15 most influential predictors identified by the proposed Optuna-optimized CatBoost classifier.

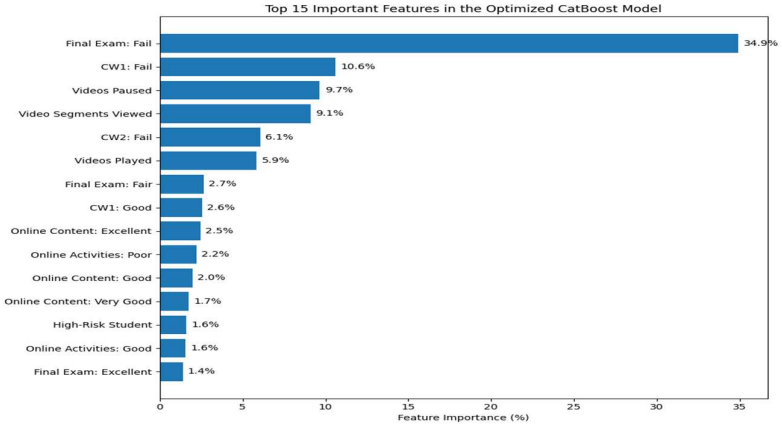


Fig. 4. Top 15 feature importance scores of the proposed optimized CatBoost model.

The results indicate that the model relied predominantly on assessment-related variables and students' online learning engagement to predict academic performance. Among all predictors, Final Exam (Fail) was by far the most influential feature,

accounting for 34.94% of the total feature importance. The next most important predictors were CW1 (Fail) (10.61%), Paused (9.66%), Segment (9.13%), CW2 (Fail) (6.07%), and Played (5.86%). Collectively, these six variables contributed approximately 76.3% of the model's total predictive importance, indicating that a relatively small subset of academic and behavioral variables contained most of the information required to distinguish between students who pass and those who fail.

The feature importance rankings further reveal that assessment outcomes constituted the strongest predictors of academic performance. In particular, the dominant contribution of Final Exam (Fail), together with the substantial importance of CW1 (Fail) and CW2 (Fail), suggests that students' performance in formative and summative assessments provides the most reliable indicators of their final academic outcome. In addition, online engagement variables, including Paused, Segment, and Played, ranked among the most influential predictors, demonstrating that students' interactions with digital learning resources also play a critical role in predicting academic success. By contrast, demographic, academic background, and institutional risk indicators contributed comparatively less to the final predictions, implying that recent assessment performance and learning behaviors provide more informative signals for early student performance prediction.

4.5. SHAP Summary Analysis

Fig. 5 exhibits the SHAP summary plot for the proposed Optuna-optimized CatBoost model, providing a global explanation of how individual features influence the prediction of student academic performance. Features are ordered based on their mean absolute SHAP values, which represent their overall contribution to the model's predictions across all observations. Consistent with the CatBoost feature importance analysis, Final Exam (Fail) ranked as the most important feature, followed by CW1 (Fail), Played, Paused, and Segment, demonstrating the stability of the feature rankings across different explainability techniques. Unlike conventional feature importance measures, the SHAP summary plot simultaneously illustrates both the magnitude and direction of each feature's contribution to the model output. Each point represents an individual student, while the color indicates the corresponding feature value, with red representing higher values and blue representing lower values. Since the model predicts Pass as the positive class, positive SHAP values increase the probability of predicting student success, whereas negative SHAP values increase the probability of predicting student failure.

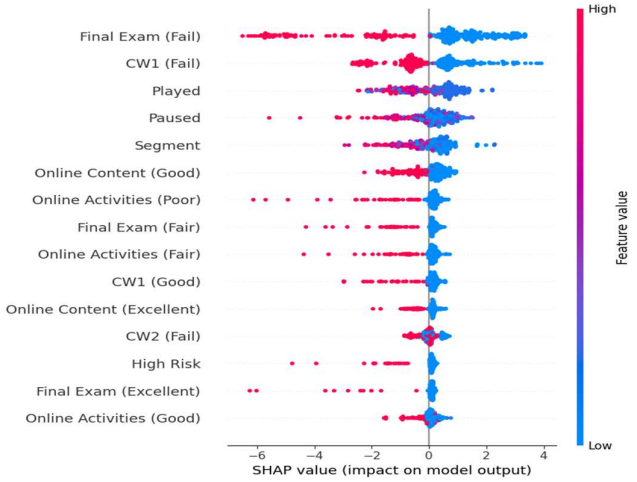


Fig. 5. SHAP summary plot of the proposed optimized CatBoost model. Features are ordered according to their mean absolute SHAP values. Each point reflects an individual student. The horizontal position indicates each feature's contribution to the predicted outcome, while the color denotes the feature value (red = high, blue = low).

For Final Exam (Fail) and CW1 (Fail), larger feature values (red points) are predominantly distributed on the negative side of the SHAP axis, indicating that students who received a failing grade in these assessments strongly contribute toward predictions of failure. Conversely, lower values of these variables (blue points), corresponding to students who did not fail these assessments, predominantly appear on the positive side of the SHAP axis, increasing the likelihood of predicting successful course completion. The engagement variables Played, Paused, and Segment also exhibit substantial SHAP value distributions, confirming that students' interactions with online learning resources meaningfully influence the model's predictions. Higher engagement generally tends to contribute positively to the prediction of student success, although the wider spread of SHAP values indicates that the effect varies across individual students because of interactions with other academic and behavioral characteristics. Similarly, variables describing online learning behavior, including Online Content (Good) and Online Activities (Poor), provide additional predictive information beyond assessment outcomes, although their overall influence is smaller than that of examination-related variables.

Overall, the SHAP summary analysis demonstrates that the proposed Optuna-optimized CatBoost model bases its predictions primarily on students' assessment performance while simultaneously incorporating behavioral indicators derived from the

online learning environment. By quantifying both the importance of each feature and its contribution to individual predictions, SHAP substantially enhances the transparency and interpretability of the proposed explainable artificial intelligence framework. Although the overall feature rankings closely resemble those obtained from the CatBoost built-in feature importance analysis, the SHAP summary plot provides richer explanations by revealing not only which features are most influential but also whether specific feature values increase the predicted probability of passing or failing for individual students. This complementary information provides deeper insights into the model's decision-making process.

5. Results Discussion

5.1. Performance of the Proposed Framework

The proposed Optuna-optimized CatBoost framework consistently outperformed statistical learning models, conventional machine learning algorithms, homogeneous ensemble methods, and heterogeneous ensemble approaches across all evaluation metrics. The significant improvements in ROC-AUC confirmed by the Wilcoxon signed-rank test demonstrate the superior discriminative ability of the proposed framework for predicting student academic performance. These results align with prior studies reporting the effectiveness of gradient boosting techniques for educational prediction (Guevara-Reyes et al., 2025; Hasan et al., 2020; Jokhan et al., 2022). However, this study extends the existing literature by demonstrating that systematic Optuna-based hyperparameter optimization can further improve the predictive capability of CatBoost beyond that achieved by conventional ML and ensemble learning methods. Although the model was evaluated using a publicly available dataset collected from a higher education institution in Oman (Hasan et al., 2021), the proposed framework provides a generalizable methodology that can be adapted to institutional learning analytics and decision-support systems at Prince Sattam bin Abdulaziz University (PSAU) and similar higher education institutions.

5.2. Effectiveness of Mutual Information Feature Selection

The Mutual Information feature selection procedure significantly improved predictive performance while reducing the feature space from 45 encoded variables to 24 informative features. Improvements in Accuracy, ROC-AUC, Balanced Accuracy, and MCC indicate that removing redundant variables enhanced model generalization while reducing computational complexity. These findings agree with previous educational data mining studies demonstrating that feature selection improves prediction by eliminating irrelevant attributes (Guevara-Reyes et al., 2025; Hasan et al., 2020). Because Mutual Information captures both linear and nonlinear dependencies, it effectively retained the most informative academic and behavioral variables. This compact feature set also facilitates practical deployment of the proposed framework within institutional learning analytics environments.

5.3. Importance of Academic and Learning Engagement Variables

Both the CatBoost feature importance analysis and SHAP explanations consistently identified assessment outcomes and learning engagement variables as the primary determinants of student performance. Final Exam (Fail), CW1 (Fail), Played, Paused, and Segment emerged as the most influential predictors, highlighting the complementary roles of academic achievement and online learning behavior. These findings are consistent with previous studies identifying assessment performance as the strongest predictor of academic success (George et al., 2025; Gori et al., 2025), while supporting learning analytics research demonstrating that students' interactions with digital learning environments provide valuable complementary information (Nawang et al., 2021; Pardo et al., 2017). The results suggest that combining academic assessments with learning engagement indicators can improve early identification of academically at-risk students.

5.4. Explainability and Institutional Decision Support

In addition to its predictive capability, the proposed framework promotes model transparency by the integration of SHAP-based explainability. While CatBoost feature importance identifies the relative importance of predictors, SHAP explains how individual feature values influence each prediction, thereby providing interpretable explanations for student success or failure. These findings support recent studies emphasizing the importance of XAI in educational data mining (George et al., 2025; Gori et al., 2025; Hidayatulloh et al., 2026). Although the framework was validated using an external educational dataset, its explainable architecture makes it suitable for institutional adoption. Within PSAU, similar learning analytics data could be integrated into the proposed framework to support transparent AI-assisted academic advising, early warning systems, and evidence-based student support.

5.5. Implications for Institutional Decision Support

The proposed framework has important implications for institutional decision support. By integrating academic performance indicators with learning engagement data, the framework enables earlier identification of students at academic risk than reliance on assessment outcomes alone. Moreover, the SHAP explanations provide actionable insights by identifying the academic and behavioral factors contributing to each prediction, allowing instructors and academic advisors to design more

targeted interventions. Although the empirical evaluation was conducted using a publicly available dataset from a higher education institution in Oman, the proposed Explainable AI framework is not institution-specific. Rather, it provides a transferable methodological framework that can be implemented at Prince Sattam bin Abdulaziz University (PSAU) using its own student information and learning management system data to support institutional learning analytics and evidence-based academic decision making.

6. Conclusion

This study proposed an XAI framework for predicting student academic performance using an Optuna-optimized CatBoost classifier. The proposed framework integrates multimodal educational data from the SIS, Moodle LMS, and eDify mobile learning application, together with Mutual Information feature selection, systematic hyperparameter optimization, and SHAP-based explainability to provide an accurate and interpretable prediction model. The empirical results demonstrate that the proposed model consistently outperformed statistical learning methods, conventional machine learning algorithms, homogeneous ensemble models, and heterogeneous ensemble learning approaches across multiple classification metrics, with the statistical significance of the observed improvements verified by the Wilcoxon signed-rank test. Mutual Information feature selection further enhanced predictive performance while reducing model complexity. The explainability analysis identified assessment outcomes, particularly Final Exam (Fail), CW1 (Fail), and CW2 (Fail), together with learning engagement indicators such as Played, Paused, and Segment, as the most influential predictors of student performance.

These findings contribute to the educational data mining and Explainable AI literature by demonstrating the effectiveness of combining Optuna-based hyperparameter optimization with CatBoost and SHAP within a unified predictive framework. The study also provides a rigorous evaluation through multiple complementary performance metrics and statistical significance testing, offering an accurate, robust, and interpretable approach for student performance prediction. From a practical perspective, the proposed framework provides a transferable decision-support tool that can assist higher education institutions in identifying academically at-risk students and supporting timely, evidence-based interventions. Although the empirical evaluation was conducted using a publicly available dataset from a higher education institution in Oman, the framework offers a conceptual foundation for developing explainable early-warning systems at institutions such as Prince Sattam bin Abdulaziz University (PSAU). Following retraining and validation using institutional data, the framework could support academic advisors, instructors, and university administrators in monitoring student progress and prioritizing academic support.

Several limitations should be acknowledged. The proposed framework was validated using a single publicly available dataset, which may restrict the generalizability of the results to different educational contexts. Furthermore, the dataset does not include psychological, socioeconomic, or social interaction variables that may further improve prediction performance. Future research should validate the framework using institutional datasets from multiple universities, investigate additional multimodal data sources, explore temporal deep learning approaches, and integrate advanced explainability techniques, such as counterfactual explanations, to further strengthen personalized educational decision support.

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Data Availability Statement

The dataset is available from the author upon reasonable request.

Conflicts of Interest

The author has no relevant financial or non-financial interests to disclose.

References

Afzaal, M., Nouri, J., Zia, A., Papapetrou, P., Fors, U., Wu, Y., Li, X., & Weegar, R. (2021). Explainable AI for data-driven feedback and intelligent action recommendations to support students self-regulation. *Frontiers in Artificial Intelligence*, *4*, 723447.

Ahmed, W., Wani, M. A., Plawiak, P., Meshoul, S., Mahmoud, A., & Hammad, M. (2025). Machine learning-based academic performance prediction with explainability for enhanced decision-making in educational institutions. *Scientific Reports*, *15*(1), 26879.

Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A next-generation hyperparameter optimization framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2623–2631.

Alamri, R., & Alharbi, B. (2021). Explainable student performance prediction models: a systematic review. *Ieee Access*, *9*, 33132–33143.

Ali, G. M. (2026a). Digital Asset Analytics for DeFi Protocol Valuation: An Explainable Optuna-Tuned Super Learner Ensemble Framework. *Journal of Risk and Financial Management*, *19*(1), 63.

Ali, G. M. (2026b). Enhancing project financial performance prediction: An explainable machine learning framework integrating frontier efficiency and super learner. *Journal of Project Management (Canada)*, *11*(1), 151–168. https://doi.org/10.5267/j.jpm.2025.10.003

Ali, G. M. (2026c). Estimating project cost of equity using explainable ensemble learning: An empirical assessment of annual report readability. *Journal of Project Management (Canada)*, *11*(2), 541–560. https://doi.org/10.5267/J.JPM.2025.12.006

Ali, G. M. (2026d). Explainable ensemble machine learning for disclosure-informed credit risk assessment in peer-to-business lending. *International Journal of Data and Network Science*, *10*(3), 1031–1048. https://doi.org/10.5267/j.ijdns.2026.4.022

Ali, G. M. (2026e). Firm valuation using accounting-based capital structure and cash holdings: An explainable machine learning approach. *International Journal of Data and Network Science*, *10*(2), 577–596. https://doi.org/10.5267/j.ijdns.2026.2.001

Ali, G. M., & Alaskar, M. Z. (2026). Nonlinear Association Between Controlling Shareholders and Financial Reporting Integrity: An Explainable Optuna-Optimized Ensemble Learning Approach in Egypt and Saudi Arabia. *Journal of Risk and Financial Management 2026, Vol. 19, Page 356*, *19*(5), 356. https://doi.org/10.3390/JRFM19050356

Ali, G., & Yahia, Z. (2025). Disclosure Determinants of Blockchain Crowdfunding Performance for Sustainable Smart City Financing: An Explainable Optuna-Optimized Machine Learning Approach. *Journal of Posthumanism*, *5*(7), 802–826. https://doi.org/10.63332/joph.v5i7.2843

Alonso, J. M., & Casalino, G. (2019). Explainable artificial intelligence for human-centric data analysis in virtual learning environments. *International Workshop on Higher Education Learning Methodologies and Technologies Online*, 125–138.

Chitti, M., Chitti, P., & Jayabalan, M. (2020). Need for interpretable student performance prediction. *2020 13th International Conference on Developments in ESystems Engineering (DeSE)*, 269–272.

Choi, W.-C., Lam, C.-T., Pang, P. C.-I., & Mendes, A. J. (2025). A systematic literature review of explainable artificial intelligence (xai) for interpreting student performance prediction in computer science and stem education. *Proceedings of the 30th ACM Conference on Innovation and Technology in Computer Science Education V. 1*, 221–227.

Cosar, M. (2024). Utilizing the eXtreme gradient boosting algorithm for artificial intelligence-supported learning analytics application. *The Eurasia Proceedings of Science Technology Engineering and Mathematics*, *28*, 277–285.

Daud, A., Aljohani, N. R., Abbasi, R. A., Lytras, M. D., Abbas, F., & Alowibdi, J. S. (2017). Predicting student performance using advanced learning analytics. *Proceedings of the 26th International Conference on World Wide Web Companion*, 415–421.

Demšar, J. (2006). Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, *7*(Jan), 1–30.

Dorogush, A. V., Ershov, V., & Gulin, A. (2018). *CatBoost: gradient boosting with categorical features support*. http://arxiv.org/abs/1810.11363

George, E. Ben, Senthilkumar, R., Al-Junaibi, F., & Al-Shuaibi, Z. (2025). Explainable AI methods for predicting student grades and improving academic success. *J. Inf. Syst. Eng. Manag*, *10*(23s), 117–126.

Gori, K. D., Zalawadia, K., & Bhaidasna, H. (2025). Explainable AI for Students Performance Prediction System. *2025 IEEE Pune Section International Conference (PuneCon)*, 1–6.

Guevara-Reyes, R., Ortiz-Garcés, I., Andrade, R., Cox-Riquetti, F., & Villegas-Ch, W. (2025). Machine learning models for academic performance prediction: interpretability and application in educational decision-making. *Frontiers in Education*, *10*, 1632315.

Hasan, R., Palaniappan, S., Mahmood, S., Abbas, A., & Sarker, K. U. (2021). Dataset of students' performance using student information system, moodle and the mobile application "eDify." *Data*, *6*(11), 110.

Hasan, R., Palaniappan, S., Mahmood, S., Abbas, A., Sarker, K. U., & Sattar, M. U. (2020). Predicting student performance in higher educational institutions using video learning analytics and data mining techniques. *Applied Sciences*, *10*(11), 3894.

Hidayatulloh, W., Mahardika, F., & Junaedi, D. I. (2026). Explainable Artificial Intelligence-Based Model for Student Academic Performance Prediction. *Journal of Information System Exploration and Research*, *4*(1), 31–40.

Hoq, M., Brusilovsky, P., & Akram, B. (2023). Analysis of an explainable student performance prediction model in an introductory programming course. *The 16th International Conference on Educational Data Mining (EDM 2023)*.

Jokhan, A., Chand, A. A., Singh, V., & Mamun, K. A. (2022). Increased digital resource consumption in higher educational institutions and the artificial intelligence role in informing decisions related to student performance. *Sustainability*, *14*(4), 2377.

Kirar, A., Veram, K., & Naga, D. (2026). A Cognitive–Generative Decision Intelligence Framework for Explainable Student Performance Prediction. *Asian Journal of Research in Computer Science*, *19*(2), 150–160.

Kunapuli, G. (2023). *Ensemble Methods for Machine Learning*. Simon and Schuster.

Lee, S., Nguyen, N., Karamanli, A., Lee, J., & Vo, T. P. (2023). Super learner machine-learning algorithms for compressive strength prediction of high performance concrete. *Structural Concrete*, *24*(2), 2208–2228.

Liu, S., & Motani, M. (2025). Improving mutual information based feature selection by boosting unique relevance. *Journal of Artificial Intelligence Research*, *82*, 1267–1292.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, *30*.

Mienye, I. D., Obaido, G., Aruleba, K., & Dada, O. A. (2022). Enhanced Prediction of Chronic Kidney Disease Using Feature Selection and Boosted Classifiers. *Lecture Notes in Networks and Systems*, *418 LNNS*, 527–537. https://doi.org/10.1007/978-3-030-96308-8_49/TABLES/4

Mohammed, A., & Kora, R. (2023). A comprehensive review on ensemble deep learning: Opportunities and challenges. In *Journal of King Saud University - Computer and Information Sciences* (Vol. 35, Number 2, pp. 757–774). King Saud bin Abdulaziz University. https://doi.org/10.1016/j.jksuci.2023.01.014

Nawang, H., Makhtar, M., & Hamzah, W. (2021). A systematic literature review on student performance predictions. *International Journal of Advanced Technology and Engineering Exploration*, *8*(84), 1441–1453.

Pardo, A., Han, F., & Ellis, R. A. (2017). Combining University student self-regulated learning indicators and engagement with online learning events to Predict Academic Performance. *IEEE Transactions on Learning Technologies*, *10*(1), 82–92. https://doi.org/10.1109/TLT.2016.2639508

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, *31*.

Sziklai, B. R., Baranyi, M., & Héberger, K. (2021). Testing Rankings with Cross-Validation. *ArXiv Preprint*. https://arxiv.org/pdf/2105.11939



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